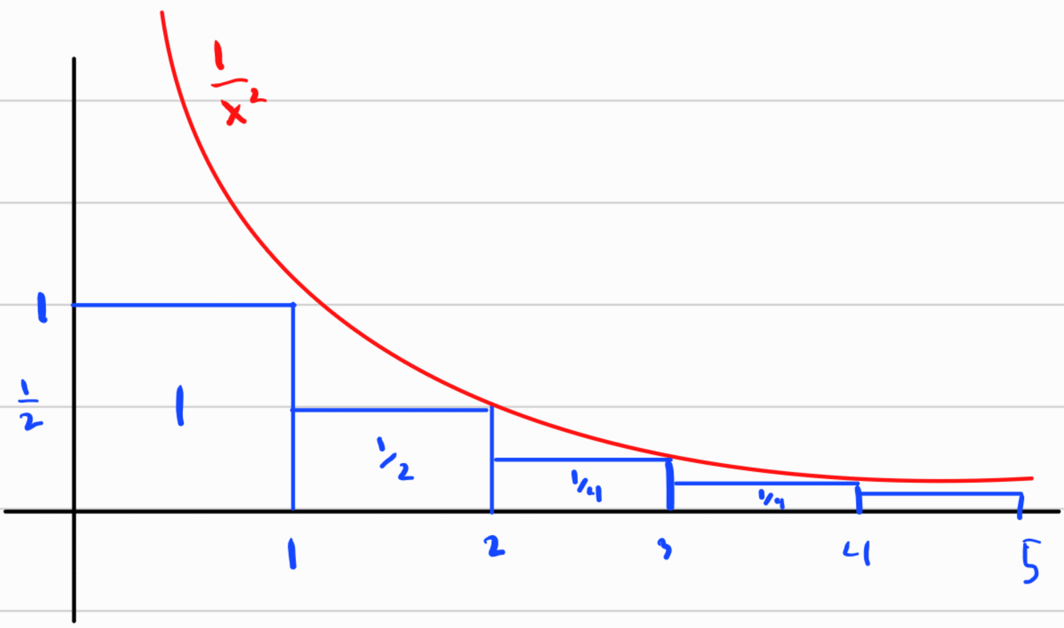
Calculating Infinite Sums is very difficult, in fact, for a general sum all we can hope to know is if is converges or diverges. So, we will spend the next sections discussing how to determine if a series converges or diverges.

The first test we look at is called the <u>integral Test</u> which relates sums to improper integrals.

As a motivating example, consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{9} + \dots$$

Now consider the function $f(x) = \frac{1}{x^2}$



All the rectangles have area equal to $\frac{1}{n^2}$, and are below the curve $\frac{1}{x^2}$. We ignore the first rectangle and see that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2} \leq 1 + \int_1^{\infty} \frac{1}{x^2}\,dx$$

The integral is improper, but we know now how to evaluate

$$\int_1^{\infty} \frac{1}{x^2}\,dx = \lim_{t\to\infty} \int_1^{t} \frac{1}{x^2}\,dx = \lim_{t\to\infty} \left.\frac{-1}{x}\right|_1^t = \lim_{t\to\infty} \frac{-1}{t} - \frac{-1}{1}$$

$$= 1$$

So

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \leq 1 + \int_1^{\infty} \frac{1}{x^2}\,dx = 2$$

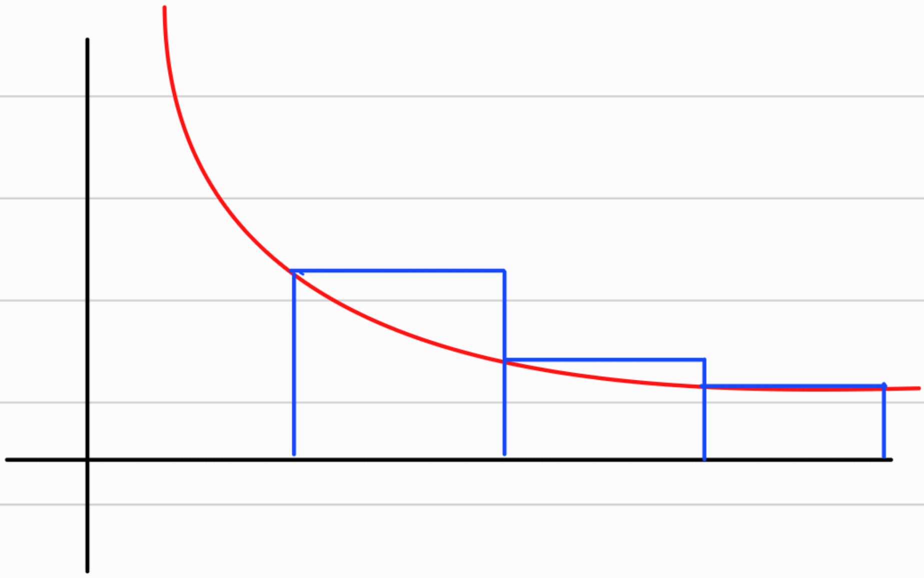
and since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is increasing (as a seq of partial sums) it must converge.

Note: Even though we have an upper bound of 2, this is a crude upper bound. The mathematician Leonhard Euler showed $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. Which is less than 2, but certainly not equal.

As another example, consider

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \quad \text{and} \quad \int_1^{\infty} \frac{1}{\sqrt{x}}\,dx$$

The set up is slightly different as the rectangles are left bound not right bound.



Here

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \geq \int_1^{\infty} \frac{1}{\sqrt{x}}\,dx$$

we know $\int_1^{\infty} \frac{1}{\sqrt{x}}\,dx = \infty$ so

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \geq \infty$$ so it diverges.

This leads us to the following integral test

The Integral Test

Suppose f is continuous, positive, and decreasing on $[1,\infty)$. Let $a_n = f(n)$ for all $n \in \mathbb{N}$. Then,

1) If $\int_1^\infty f(x)\,dx$ is convergent, then the series $\sum_{n=1}^{\infty} a_n$ is convergent as well.

2) If $\int_1^\infty f(x)\,dx$ is divergent, then the series $\sum_{n=1}^{\infty} a_n$ is divergent as well.

Note: a) The integral and sum don't need to start at 1, for $\sum_{n=4}^{\infty} \frac{1}{(n-3)^3}$ we use $\int_4^\infty \frac{1}{(x-3)^3}\,dx$ and notice it converges

b) The requirement of $f(x)$ decreasing just needs to be eventually true. So as long as $f(x)$ is decreasing for $x > N$ for some value N, it still works.

Ex: $$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

The function $f(x) = \frac{1}{x^2+1}$ is positive, decreasing, continuous, and $f(n) = \frac{1}{n^2+1}$ so we can use the integral test.

$$\int_1^{\infty} \frac{1}{x^2+1}\,dx = \lim_{t \to \infty} \arctan(x)\Big|_1^t = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

So $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ is convergent

Note: To use the integral test, we need to evaluate $\int_1^{\infty} f(x)\,dx$ so there is some limitation.

Recall that $\int_1^{\infty} \frac{1}{x^p}\,dx$ converges for $p > 1$
diverges for $p \leq 1$

Since $\frac{1}{x^p}$ is cont, dec, and positive for $0 < p$
the integral test shows

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$ converges for $p > 1$, and
diverges for $p \leq 1$ ($p = 1$ is harmonic series)

This is an important result called a P-Series

Thm:
The P-Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$

Ex:
$\sum \frac{1}{n^3}$ Converges as it is a P-Series with $p = 3 > 1$

$\sum \frac{1}{\sqrt[3]{n}}$ diverges as it is a P-Series with $p = \frac{1}{3} \leq 1$

Ex: $\sum_{n=1}^{\infty} \frac{\ln(n)}{n}$

We use $f(x) = \frac{\ln(x)}{x}$. Certainly Continuous and Positive. So we check decreasing

$f'(x) = \frac{1 - \ln(x)}{x^2}$ since we only care about the function eventually decreasing, we see $f'(x) < 0$ for $x > e$. This is fine.

Then

$$\int_1^\infty \frac{\ln(x)}{x}\,dx = \int_0^\infty u\,du = \lim_{t \to \infty} \frac{u^2}{2}\Big|_0^t = \infty - 0$$

is divergent

So

$$\sum_{n=1}^{\infty} \frac{\ln(n)}{n}$$ diverges via the Integral test.